Original Research Article

International Journal of Silkroad Institute of Research and Training (IJSIRT)

Vol-4. No. 1, Jan-June 2026 | DOI: 10.3126/ijsirt.v4i1.XXXXX | ISSN: 2990-7837 (Online & Print) |

ECG Changes in Patients with Chronic Kidney Disease: A Cross-Sectional Observational Study

Ranjit Kumar Yadav [1], **Shree Narayan Yadav** [1]

[1] *Department of Internal Medicine, Bir Hospital, Kathmandu*

Received: 19th Mar, 2026 • **Accepted:** 10th May, 2026 • **Published:** 30th June, 2026

ABSTRACT

Background: Chronic kidney disease (CKD) is often associated with cardiovascular abnormalities, including left ventricular hypertrophy (LVH), conduction disturbances, and arrhythmias. Early electrocardiographic (ECG) assessment is crucial for detection and monitoring. The objective is to evaluate ECG changes in CKD patients.

Method: A descriptive, cross-sectional study was conducted among 72 CKD patients confirmed by clinical, laboratory, and imaging criteria. Statistical analysis was performed using SPSS 25.0.

Results: Of 72 patients (68.1% male, mean age 52.32 ± 17.15 years), hypertension was the leading etiology (37.5%). ECG abnormalities were found in 55.5%, with LVH being the most common (33.3%). Anemia and electrolyte disturbances were prevalent.

Conclusion: LVH was the most frequent ECG abnormality in CKD, with hypertension and diabetes as common etiologies. Cardiovascular abnormalities were significantly higher than in the general population, highlighting the need for early intervention.

Keywords: Chronic Kidney Disease; Left Ventricular Hypertrophy; Electrocardiography.

Correspondence: Dr. Ranjit Kumar Yadav,Department of Internal Medicine, Bir Hospital, Kathmandu, E-mail: kranjit_11@yahoo.com

INTRODUCTION

Chronic kidney disease (CKD) has emerged as a major global public health concern, affecting approximately 11–13% of the population and contributing to over 850 million cases worldwide [1]. It is characterized by a progressive and often irreversible decline in glomerular filtration rate (GFR) or structural kidney damage lasting more than three months. As the disease progresses, patients face a disproportionately high burden of morbidity and mortality, primarily due to cardiovascular complications[2].

Cardiovascular disease (CVD) remains the leading cause of death among CKD patients, with over 50% of mortality in this group attributed to cardiac causes [3]. The risk of developing cardiovascular events is significantly higher in CKD patients even those in early stages compared to the general population [4]. Traditional risk factors such as hypertension, diabetes mellitus, and dyslipidemia are prevalent in CKD; however, CKD specific non traditional factors such as chronic inflammation, volume overload, vascular calcification, uremic toxins, and anemia further exacerbate the risk [5–7].

These pathophysiological mechanisms contribute to structural cardiac remodeling, particularly left ventricular hypertrophy (LVH), myocardial fibrosis, and systolic/diastolic dysfunction. Moreover, neurohormonal activation especially of the sympathetic nervous system and the renin angiotensin aldosterone system plays a crucial role in the progression of myocardial abnormalities [8]. The cumulative effect of these insults leads to various cardiac arrhythmias, ischemic changes, and conduction disturbances [9].

Electrocardiography (ECG), a widely available, inexpensive, and non-invasive diagnostic tool, plays an essential role in detecting early cardiac involvement in CKD. Common ECG abnormalities include prolonged QT intervals, LVH, ST-T wave abnormalities, and bundle branch blocks. These findings not only reflect underlying structural disease but are also predictive of adverse cardiovascular outcomes [10]. Studies have reported ECG abnormalities in up to 7% of dialysis-dependent CKD patients, with LVH being among the most prevalent [11–13].

Furthermore, electrolyte disturbances such as hyperkalemia and hypocalcemia frequently encountered in CKD can cause dynamic ECG changes that, if unrecognized, may result in fatal arrhythmias. Anemia, another common CKD complication, also contributes to myocardial hypoxia and subsequent electrical instability [14].

Despite its clinical utility, the ECG is underutilized in many resource limited settings, including South Asia, where CKD is often diagnosed at later stages and where access to advanced cardiac imaging is limited. In Nepal, where hypertension and diabetes prevalence are increasing, the cardiovascular burden among CKD patients is expected to rise. Yet, few studies have evaluated the spectrum of ECG abnormalities in this context. Therefore, this study was conducted to analyze the pattern of ECG changes in patients with CKD admitted to a tertiary care center in Nepal and to correlate these findings with demographic, clinical, and biochemical parameters.

The findings aim to enhance early recognition of cardiovascular risk in CKD, thus supporting timely and targeted intervention.

METHODS

This was a hospital-based, descriptive cross-sectional study conducted over a one-year period from August 1, 2021, to July 31, 2022, at Bir Hospital, a tertiary care and referral center in Kathmandu, Nepal. The objective of the study was to evaluate electrocardiographic (ECG) abnormalities in patients with chronic kidney disease (CKD).

The sample size was calculated using the standard formula for prevalence studies [15]:

$$n = \frac{Z^2 pq}{d^2}$$

where $Z = 1.96$ (for 95% confidence interval),
$p = 0.75$
(assumed prevalence of ECG abnormalities in CKD),
$q = 1 - p = 0.25$, and
$d = 0.10$ (allowable margin of error).
The calculated minimum sample size was 72 patients.

A non-probability purposive sampling method was employed. Eligible participants included adult patients (aged 18 years) diagnosed with CKD of any etiology or stage, including those on maintenance dialysis, admitted to the nephrology ward or emergency department. The diagnosis of CKD was confirmed using clinical, biochemical, and imaging findings, consistent with the KDIGO 2012 classification. Patients with acute myocardial infarction, structural heart disease (including congenital, valvular, or cardiomyopathic abnormalities), pregnant individuals, and those under the age of 18 were excluded.

After obtaining written informed consent, eligible patients were enrolled consecutively. Ethical clearance for the study was granted by the Institutional Review Board of Bir Hospital. Data were collected using a structured and pre-validated case record form, which included demographic variables (age, sex, socioeconomic status), CKD-related history (duration, stage, dialysis status), and comorbid conditions. Hypertension was defined as blood pressure 140/90 mmHg or use of antihypertensive medication, and diabetes mellitus was defined as fasting blood glucose 126 mg/dL, postprandial glucose 200 mg/dL, or the use of hypoglycemic agents.

Each patient underwent standard laboratory investigations including complete blood count, serum creatinine, blood urea nitrogen (BUN), and serum electrolytes (sodium, potassium, calcium). Urinalysis and abdominal-pelvic ultrasonography were performed to assess renal morphology. Estimated glomerular filtration rate (eGFR) was calculated using the CKD-EPI formula to determine CKD staging. All patients underwent a standard 12-lead ECG recorded at a paper speed of 25 mm/s and amplitude of 10 mm/mV. Electrocardiograms were interpreted independently by two trained physicians blinded to clinical data. ECG abnormalities were classified into categories: left ventricular hypertrophy (LVH), arrhythmias (e.g., atrial fibrillation, premature ventricular contractions), conduction disturbances (e.g., atrioventricular or bundle branch block), ST-T wave abnormalities (e.g., ischemic or repolarization changes), and axis deviations or atrial enlargements.

Data were entered into Microsoft Excel and analyzed using SPSS version 25.0 (IBM Corp., Armonk, NY, USA). Descriptive statistics, including mean and standard deviation for continuous variables and frequencies with percentages for categorical variables, were used to summarize patient characteristics and ECG findings. Cross-tabulations were employed to explore distributions and patterns of ECG abnormalities across CKD stages. Given the descriptive design, no inferential statistical testing was performed.

Most patients (65.3%) had hemoglobin levels between 8–9.9 g%, reflecting the high prevalence of anemia in CKD.

RESULTS

This section presents the demographic, clinical, and biochemical characteristics of the 72 patients diagnosed with chronic kidney disease (CKD) included in the study. The majority of patients (23.6%) were in the 51–60 years age group, with a mean age of 52.32 ± 17.15 years, suggesting a mid-to-late adult onset pattern. Hypertension emerged as the most common cause of CKD (37.5%), followed by combined hypertension and diabetes mellitus (29.2%). A striking 86.1% of patients were in Stage G5, indicating advanced kidney disease at presentation. Most patients (65.3%) had hemoglobin levels between 8–9.9 g%, reflecting the high prevalence of anemia in CKD. The predominant serum creatinine range was 8–10 mg/dL (40.3%), highlighting marked renal function decline. ECG abnormalities were observed in 55.5% of patients, with left ventricular hypertrophy (33.3%) being the most frequent finding.

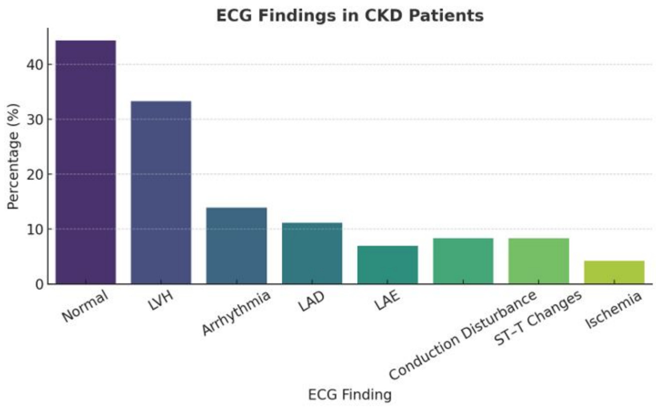

Figure 1. Etiology of CKD

Figure 1 shows hypertension was the most prevalent cause of CKD, followed by combined hypertension and diabetes mellitus.

Table 1. Distribution of Study Participants by Demographic and Clinical Characteristics

Characteristic	Frequency	Percentage (%)
Age Group (Years)		
21–30	10	13.9
31–40	15	20.8
41–50	6	8.3
51–60	17	23.6
61–70	11	15.3
71–80	11	15.3
81–90	1	1.4
91–100	1	1.4
Etiology		
Hypertension	27	37.5
Diabetes Mellitus	8	11.1
Hypertension + Diabetes	21	29.2
Others	16	22.2
CKD Stage		
G1	0	0.0
G2	0	0.0
G3a	0	0.0
G3b	0	0.0
G4	10	13.9
G5	62	86.1
Hb Level (g%)		
< 6	2	2.8
6–7.9	14	19.4
8–9.9	47	65.3
10–11.9	9	12.5
Urea (mg/dL)		
50–99	16	22.2
100–149	46	63.9
150–199	7	9.7
≥ 200	3	4.2
Creatinine (mg/dL)		
4–5.9	15	20.8
6–7.9	23	31.9
8–10	29	40.3
> 10	5	6.9
ECG Finding		
Normal	32	44.4
LVH	24	33.3
Arrhythmia	10	13.9
LAD	8	11.1
LAE	5	6.9
Conduction Disturbance	6	8.3
ST–T Changes	6	8.3
Ischemia	3	4.2

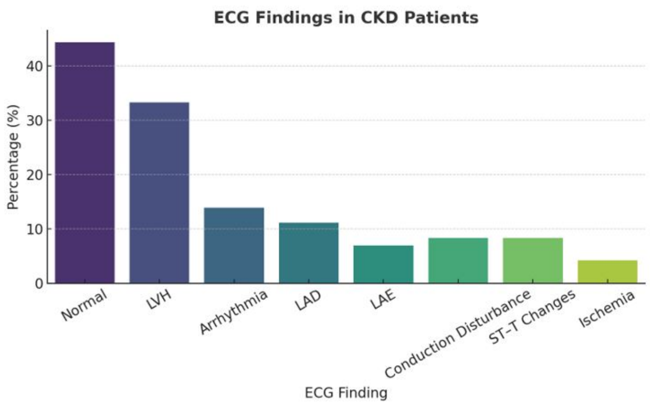

Figure 2. CKD Staging

Figure 2 shows the vast majority of patients were in stage G5 at presentation, indicating late detection or progression.

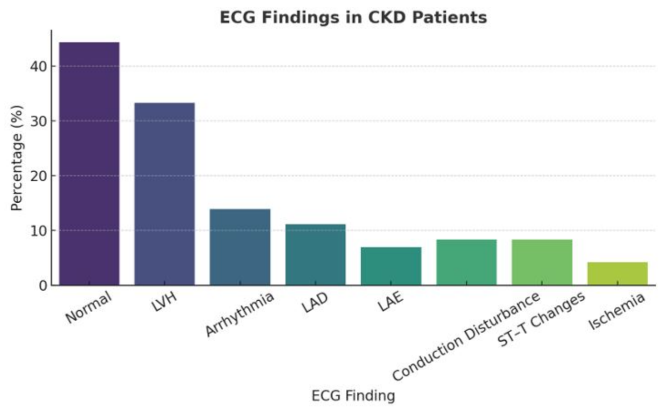

Figure 3. Hemoglobin Levels

Figure 3 shows that most patients exhibited moderate anemia, with hemoglobin levels concentrated in the 8–9.9 g% range.

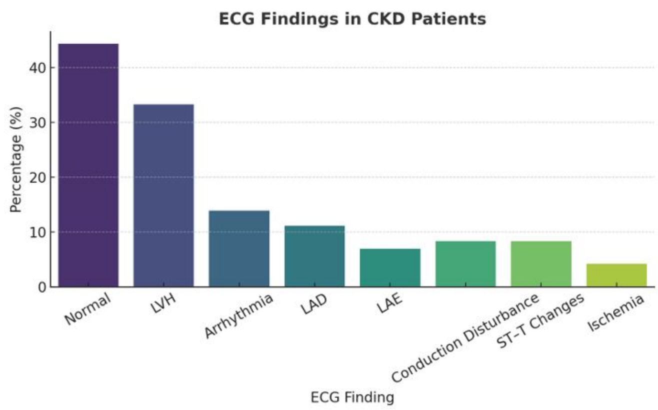

Figure 4. ECG Findings

Figure 4 show that more than half of the patients showed ECG abnormalities, particularly left ventricular hypertrophy.

DISCUSSIONS

Cardiovascular disease (CVD) is a well-established and often underrecognized complication of chronic kidney disease (CKD). Accelerated atherosclerosis, left ventricular hypertrophy (LVH), arrhythmias, and myocardial fibrosis are all part of the complex interplay between renal dysfunction and cardiac remodeling. CKD promotes hypertension, volume overload, dyslipidemia, and anemia, all of which contribute to increased cardiac workload and structural damage. These changes are frequently detectable by electrocardiography (ECG), a low-cost and accessible diagnostic modality that plays a crucial role in cardiovascular risk assessment in CKD patients.

In the present study involving 72 CKD patients, ECG abnormalities were identified in 55.5% of patients. This prevalence is slightly lower than that reported in some prior studies, including a study by Joshi et al., where 77.1% of CKD patients exhibited ECG abnormalities [16]. The difference may reflect variations in sample size, dialysis status, or population characteristics.

The most common ECG finding in our study was left ventricular hypertrophy (LVH), observed in 33.3% of patients. This finding is in agreement with studies by Krivoshiev et al. (33.6%) [13] and Joshi et al. (23%)[16]. Soman et al. reported a slightly lower prevalence of 18% in a smaller cohort [17]. LVH in CKD is driven by chronic pressure overload, anemia, and activation of the renin-angiotensin-aldosterone system (RAAS), resulting in concentric hypertrophy and eventual diastolic dysfunction.

Arrhythmias were observed in 13.9% of the patients in our study. Soman et al. and Goornavar et al. reported prevalence rates of 5% and 16%, respectively[17, 18]. Arrhythmias in CKD are often precipitated by electrolyte imbalances—particularly hyperkalemia and hypocalcemia—as well as myocardial fibrosis and autonomic dysfunction. Given the strong link between arrhythmias and sudden cardiac death in CKD, early identification and management are critical.

Conduction disturbances were present in 8.3% of patients in this study. This figure is slightly lower than those reported by Soman et al. (15%), Krivoshiev et al. (10.9%), and Joshi et al. (11.4%) [13, 16, 17]. Conduction defects in CKD often arise from uremic toxins and fibrotic degeneration of conduction pathways, leading to bundle branch blocks or atrioventricular conduction delays.

Other abnormalities in our study included left axis deviation (LAD) (11.1%) and left atrial enlargement (LAE) (6.9%). These findings are consistent with those reported by Joshi et al. (11.4% and 2.9%, respectively) [16]. Axis deviation and atrial enlargement reflect chronic pressure overload, fluid retention, and structural remodeling due to long-standing hypertension or volume expansion, common in late-stage CKD.

Ischemic changes, seen in 4.2% of our patients, were lower than in earlier reports by Soman et al. (32%) and Krivoshiev et al. (29.1%) [13, 17]. This discrepancy may stem from the use of more stringent ECG criteria or the exclusion of patients with known ischemic heart disease in our study. Nonetheless, ischemic ECG patterns in CKD are significant, as silent myocardial ischemia is common due to uremic neuropathy and altered pain perception.

The mean hemoglobin level in our cohort was 8.51 ± 1.18 g/dL, with the majority falling below 10 g/dL. Anemia is a key contributor to LVH and reduced myocardial oxygen delivery in CKD. These findings are consistent with data from Singh et al. (6.7 g/dL), Chafekar et al. (7.84 g/dL), and Foley et al. (8.4 g/dL) [11, 19, 20].

The mean serum creatinine and urea levels were 7.43 ± 2.51 mg/dL and 120.48 ± 35.92 mg/dL, respectively. These elevated values highlight the advanced renal dysfunction in our sample. Studies by Singh et al. and Foley et al. report similar trends in uremia and creatinine burden [11, 19].

Although our ECG abnormality prevalence is slightly lower than in other cohorts, the results underscore the significant burden of subclinical cardiac abnormalities in CKD. It is notable that even in the absence of overt symptoms, nearly half of the patients had clinically relevant ECG findings. Routine ECG screening in CKD patients, particularly in stages G4 and G5, may offer substantial clinical value by identifying early cardiac changes and guiding timely referral for cardiology evaluation or echocardiography.

The relatively high frequency of LVH, arrhythmias, and conduction abnormalities in this study supports the need for multidisciplinary care involving both nephrologists and cardiologists. Given the resource limitations in many developing settings, ECG serves as a cost-effective frontline tool for cardiovascular screening.

This study has several limitations. It is a single-center study with a relatively small sample size and may not reflect community-level trends. The absence of echocardiographic correlation limits assessment of structural changes and ejection fraction. Additionally, the cross-sectional design precludes analysis of causal relationships or progression.

Future studies with larger, multicentric cohorts and incorporation of echocardiography, cardiac biomarkers, and long-term cardiovascular outcomes would provide a more holistic understanding of ECG changes in CKD.

CONCLUSIONS

Left ventricular hypertrophy was the most common ECG abnormality in CKD patients, followed by conduction disturbances, arrhythmias, and ischemic changes. Hypertension and diabetes were the primary etiological factors, highlighting the need for early screening to prevent cardiovascular complications and CKD progression. Anemia and electrolyte imbalances were prevalent and require regular monitoring, as they contribute to cardiac dysfunction. Early interventions, including effective control of hypertension and anemia, are essential to reduce cardiovascular risks in CKD patients.

Conflict of Interest: None

Funding: None

REFERENCES

[1] Nathan R. Hill, Samuel T. Fatoba, Jason L. Oke, Jennifer A. Hirst, Christopher A. O'Callaghan, Daniel S. Lasserson, and F. D. Richard Hobbs. Global prevalence of chronic kidney disease – a systematic review and meta-analysis. *PLoS ONE*, 11(7):e0158765, 2016. doi:10.1371/journal.pone.0158765.

[2] Ernesto L. Schiffrin, Mark L. Lipman, and Johannes F. E. Mann. Chronic kidney disease: Effects on the cardiovascular system. *Circulation*, 116(1):85–97, 2007. doi:10.1161/CIRCULATIONAHA.106.678342.

[3] Mark J. Sarnak, Andrew S. Levey, Anton C. Schoolwerth, Josef Coresh, Bruce Culleton, L. Lee Hamm, Peter A. McCullough, Bertram L. Kasiske, Ellie Kelepouris, Michael J. Klag, Patrick Parfrey, Marc Pfeffer, Leopoldo Raij, David J. Spinosa, and Peter W. Wilson. Kidney disease as a risk factor for development of cardiovascular disease: A statement from the american heart association councils on kidney in cardiovascular disease, high blood pressure research, clinical cardiology, and epidemiology and prevention. *Circulation*, 108(17):2154–2169, 2003. doi:10.1161/01.CIR.0000095676.90936.80.

[4] Guruprasad Manjunath, Hocine Tighiouart, Hassan Ibrahim, Bonnie MacLeod, Deeb N. Salem, John L. Griffith, Josef Coresh, Andrew S. Levey, and Mark J. Sarnak. Level of kidney function as a risk factor for atherosclerotic cardiovascular outcomes in the community. *Journal of the American College of Cardiology*, 41(1):47–55, 2003. doi:10.1016/S0735-1097(02)02663-3.

[5] Gerard M. London, Sylvain J. Marchais, Alain P. Guerin, and Fabien Metivier. Arteriosclerosis, vascular calcifications and cardiovascular disease in uremia. *Current Opinion in Nephrology and Hypertension*, 14(6):525–531, 2005. doi:10.1097/01.mnh.0000168336.67499.c0.

[6] Yen-Wen Liu, Chi-Ting Su, Yu-Tzu Chang, Wei-Chuan Tsai, Yu-Ru Su, Saprina P. H. Wang, Chun-Shin Yang, Liang-Miin Tsai, Jyh-Hong Chen, and Junne-Ming Sung. Elevated serum interleukin-18 level is associated with all-cause mortality in stable hemodialysis patients independently of cardiac dysfunction. *PLoS ONE*, 9(3):e89457, 2014. doi:10.1371/journal.pone.0089457.

[7] Paul Muntner, Suzanne E. Judd, Liyan Gao, Orlando M. Gutiérrez, et al. Cardiovascular risk factors in ckd associate with both esrd and mortality. *Journal of the American Society of Nephrology*, 24 (7):1159–1165, 2013. doi:10.1681/ASN.2012070642.

[8] J. Neumann, G. Ligtenberg, I. I. Klein, et al. Sympathetic hyperactivity in chronic kidney disease: Pathogenesis, clinical relevance, and treatment. *Kidney International*, 65(5):1568–1576, 2004. doi:10.1111/j.1523-1755.2004.00552.x.

[9] Marian Goicoechea, Soledad García de Vinuesa, Francisco Gómez-Campderá, and José Luño. Predictive cardiovascular risk factors in patients with chronic kidney disease. *Kidney International*, 67 (Supplement 93):S35–S38, 2005. doi:10.1111/j.1523-1755.2005.09308.x.

[10] Rajan Sharma, David C. Gaze, David Pellerin, Rajnikant L. Mehta, Helen Gregson, Christopher P. Streather, Peter O. Collinson, and Stephen J. D. Brecker. Cardiac structural and functional abnormalities in end stage renal disease patients with elevated cardiac troponin t. *Heart*, 92(6):804–809, 2006. doi:10.1136/hrt.2005.069666.

[11] Robert N. Foley, Patrick S. Parfrey, John D. Harnett, Gerald M. Kent, Christopher J. Martin, David C. Murray, and Paul E. Barre. Clinical and echocardiographic disease in patients starting end-stage renal disease therapy. *Kidney International*, 47(1):186–192, 1995. doi:10.1038/ki.1995.22.

[12] M. Dahan, P. Siohan, B. Viron, C. Michel, C. Paillole, R. Gourgon, and F. Mignon. Relationship between left ventricular hypertrophy, myocardial contractility, and load conditions in hemodialysis patients: An echocardiographic study. *American Journal of Kidney Diseases*, 30(6):780–785, 1997. doi:10.1016/S0272-6386(97)90082-2.

[13] S. Krivoshiev, Z. Kiriakov, and S. Antonov. Electrocardiographic changes in patients with chronic kidney failure treated by periodic hemodialysis. *Vutreshni Bolesti*, 26(5):64–67, 1987.

[14] Gerard M. London, F. Fabiani, Sylvain J. Marchais, Marie-Christine de Vernejoul, Alain P. Guerin, Michel E. Safar, Fabien Metivier, and F. Llach. Uremic cardiomyopathy: An inadequate left ventricular hypertrophy. *Kidney International*, 31(4):973–980, 1987. doi:10.1038/ki.1987.94.

[15] Bijay Lal Pradhan. *Sampling Techniques Using R*, volume 1. Silkroad Institute of Research and Training, Kathmandu, Nepal, 2025.

[16] Pankaj Joshi. Study of clinical, electrocardiographic and 2d-echocardiography findings in chronic kidney disease patients. *IOSR Journal of Dental and Medical Sciences*, 21(1):51–54, 2022. doi:10.9790/0853-2101055184.

[17] Sandeep S. Soman, Keisha R. Sandberg, Steven Borzak, Michael P. Hudson, Jerry Yee, and Peter A. McCullough. The independent association of renal dysfunction and arrhythmias in critically ill patients. *Chest*, 122(2):669–677, 2002. doi:10.1378/chest.122.2.669.

[18] S. M. Goornavar, R. Pramila Devi, and R. M. Ashoka. A study of echocardiographic changes in patients with chronic kidney disease. *Medica Innovatica*, 4(1):1–5, 2015.

[19] N. P. Singh, Chandrashekhar, M. Nair, S. Anuradha, R. Kohli, and S. K. Agarwal. The cardiovascular and hemodynamic effects of erythropoietin in chronic renal failure. *Journal of the Association of Physicians of India*, 48(3):301–306, 2000.

[20] S. A. Kale, N. S. Kulkarni, S. Gang, A. Ganju, L. Shah, and M. M. Rajapurkar. Left ventricular disorders in patients of end stage renal disease entering hemodialysis programme. *Indian Journal of Nephrology*, 11:12–16, 2001. doi:10.4103/0971-4065.34808.

Citation: Yadav R K, Yadav S N. ECG Changes in Patients with Chronic Kidney Disease: A Cross-Sectional Observational Study. *IJSIRT*. 2026;4(1):22–27. DOI: 10.3126/ijsirt.v4i1.XXXXX